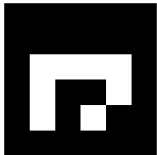


ArUco DICT_4X4_50 — 20 mm

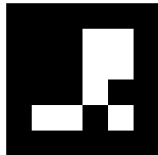
Print at 100%. Turn OFF 'fit to page' / 'shrink oversized pages'.
Measure the black square with a ruler before use: it must read 20 mm across.



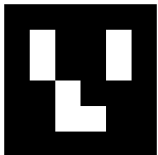
ID 0 · 20 mm



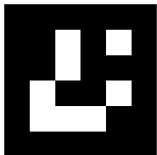
ID 1 · 20 mm



ID 2 · 20 mm



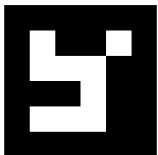
ID 3 · 20 mm



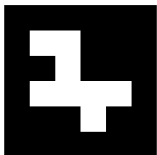
ID 4 · 20 mm



ID 5 · 20 mm



ID 6 · 20 mm



ID 7 · 20 mm

촬영 조건표 · Shooting grid

One marker per container. Note the container's diameter: it is the variable that decides angle tolerance, so a shot without it cannot be placed on the curve.

Include glossy containers. Specular highlight is the factor the synthetic study could only bound, and matte-only photographs would leave it exactly where it was.

Shoot each scene at two exposures. The model predicts detection turns over on the highlight's peak brightness rather than the light's angular size, so exposure is the first prediction worth testing.

Vary	Levels
Container	지름 기록 · 유광/무광 구분 · 5~10종
Distance	30 cm / 60 cm / 1 m
Angle	0° / 15° / 30° / 45° / 60°
Lighting	실내 확산광 / 직사 조명 (정반사 유도)
Exposure	자동 노출 / 하이라이트 기준 -2 EV

60 frames per container (3 x 5 x 2 x 2). Eight containers gives 480.

Halve it by dropping to one lighting condition if time runs short — but keep both exposures.

Record with every frame

container ID, container diameter (mm), surface (glossy / matte), marker ID, marker size (20 mm), distance, angle, lighting, exposure.

A frame whose conditions were not written down is not a measurement.